

Problem #1

```
# Basinger, Trueman, Duran, Druce-Hoffman, AerE 1610, Python Lab 1
# Check if a number is even or odd
print('check if number is odd or even')

number = int(input("enter a number: "))

if number == 0:
    print('number is zero')
elif number % 2 == 0:
    print(f"Number {number} is even")
else:
    print(f"Number {number} is odd")

print("thanks.")
```

Output :

```
check if number is odd or even
enter a number: 3
Number 3 is odd
thanks.
```

Problem #2

```
print(12345.6789)  
12345.6789
```

```
print(f'{12345.6789:10.4f}')  
12345.6789
```

```
print(f'{12345.6789:2.2f}')  
12345.68
```

Problem #3

```
import numpy as np # import numpy library

x = int(input('Enter integer x: ')) # prompt user for x
t = int(input('Enter integer t: ')) # prompt user for t

z = (9*np.pi*t + 10*np.cos(x)) / (np.sqrt(t) - abs(np.sin(t))) * np.exp(x) # calculate z

print('z = %.2f' % z) # print z with 2 decimal places
```

Output :

```
Enter integer x: 10
Enter integer t: 1
z = 2762685.71
```

Problem #4

```
n = input("Enter an integer number n:") #get num
m = n+n #get num 1 in str
o = n+n+n #get num 2 in str
p = int(n)+int(m)+int(o) #get num final in num
print(f"the n+nn+nnn={p}") #print
```

Output :

Enter an integer number n:2
the n+nn+nnn=246

Enter an integer number n:5
the n+nn+nnn=615

Problem #5

```
import numpy as np # import numpy library

# print menu options
print('Menu') # print menu
print(' 1. Cylinder') # print option 1 as cylinder
print(' 2.Circle') # print option 2 as circle
print(' 3.Rectangle') # print option 3 as rectangle

choice = int(input('Please choose one: ')) # prompt user for choice

if choice == 1: # cylinder # check if the user enters 1
    r = float(input('Enter the radius of the cylinder in in.: ')) # prompt for radius
    h = float(input('Enter the height of the cylinder in in.: ')) # prompt for height
    area = 2 * np.pi * r * h + 2 * np.pi * r**2 # calculate surface area
    print('The area is %.2f in^2' % area) # print area

elif choice == 2: # check if user enters 2
    r = float(input('Enter the radius of the circle in in.: ')) # prompt for radius
    area = np.pi * r**2 # calculate area
    print('The area is %.2f in^2' % area) # print area

elif choice == 3: # rectangle # check if the user enters 3
```

```
l = float(input('Enter the length in in: ')) # prompt for length
w = float(input('Enter the width in in: ')) # prompt for width
area = l * w # calculate area
print('The area is %.2f in^2' % area) # print area

else: # invalid choice
    print('Invalid choice') # print error message
```

Output:

Menu

1. Cylinder

2.Circle

3.Rectangle

Please choose one: 1

Enter the radius of the cylinder in in.: 8

Enter the height of the cylinder in in.: 9

The area is 854.51 in^2

Menu

1. Cylinder

2.Circle

3.Rectangle

Please choose one: 2

Enter the radius of the circle in in.: 6

The area is 113.10 in^2

Menu

1. Cylinder

2.Circle

3.Rectangle

Please choose one: 3

Enter the length in in: 9

Enter the width in in: 5

The area is 45.00 in^2